

MaxwellFD: 2D Yee FDTD and FDFD

Physics, methods, and code

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Abstract

MaxwellFD is a Python Yee-cell solver. One uniform Cartesian grid, one material sampling, and one pair of discrete curls feed both a leapfrog (FDTD) and a sparse frequency-domain system (FDFD). Phasors use $e^{j\omega t}$, the same convention as MaxwellMOM. This note describes the physics and the method that are actually coded, how the package is organized, and how the Fresnel-slab, dielectric-cylinder, PML line-current, and open PEC-cylinder examples are run and drawn. The equation reference is `FD_LaTeX_Reference.tex`. The convolutional PML from that note is coded as a stretch of the normal derivatives on a PEC-backed band. Embedded PEC and PMC objects remove Yee samples, and an open PEC cylinder is compared with the Mie series on that band. An open dielectric cylinder on the same band is driven by the contrast current $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$. A 3D PEC cavity checks three pure modes of the Yee cell. The resistive sheet and the GSTC cell remain specified in that note.

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1 Status

Version 0.1.0 contains the following.

1. The 2D cell in free space and in a uniform isotropic medium: TMz and TEz curls, PEC and periodic walls, the CFL limit, cavity eigenvalues, a manufactured FDFD solve, and a periodic standing wave.
2. Volumetric media. A staircase assigns a real ε , μ , and $\sigma \geq 0$ to each Yee sample. FDFD accepts a complex $\varepsilon(\omega)$. FDTD adds one bulk Debye pole or one bulk Lorentz pole through a polarization current. The closed-box checks are a Fresnel slab and a dielectric cylinder, both driven by an analytic Dirichlet trace on the existing PEC wall.
3. The convolutional PML on a PEC-backed band, for both polarizations and both drivers. In FDFD every derivative carries the factor $1/s_w(\omega)$. The leapfrog keeps one auxiliary for each of those derivatives. A line current is compared with the outgoing Hankel function on the unstretched samples.
4. Embedded PEC and PMC objects. PEC drops every electric sample inside its region. PMC drops every magnetic sample. The open check is a PEC cylinder: the unknown is the scattered field, the object carries the trace $-E_{\text{inc}}$, and the PML absorbs the outgoing wave.
5. An open dielectric cylinder on that PML. The staircase disk stays in the operator. The incident wave is the impressed current $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$, and a free sample of the total field is $E_s + E_{\text{inc}}$.
6. The 3D Yee cell and a sparse FDFD operator on a rectangular box. Uniform isotropic media, with PEC and periodic walls. Three pure cavity modes are checked by a Rayleigh quotient against the semi-discrete frequency with the third centered difference.

MaxwellFD does not import MaxwellMOM. The shared pieces are the time convention, the four SI constants, and the analytic targets used as external checks. The 3D grid is a rectangular box with a uniform isotropic medium. Sheets, a 3D PML, and a 3D leapfrog remain later slices. Embedded PEC and PMC objects remove Yee samples on the 2D grid. The open PEC cylinder and the open dielectric cylinder are scattered-field solves on the PML.

2 Physics

Fields are in SI units. With the convention $e^{j\omega t}$, a wave $e^{-jk\hat{\mathbf{k}}\cdot\mathbf{r}}$ travels in the direction $\hat{\mathbf{k}}$, and ∂_t becomes $j\omega$. The constants in `maxwell_fd.utils.constants` are

$$\begin{aligned} c_0 &= 299792458 \text{ m s}^{-1}, & \varepsilon_0 &= 8.8541878128 \times 10^{-12} \text{ F m}^{-1}, \\ \mu_0 &= 1.25663706212 \times 10^{-6} \text{ H m}^{-1}, & \eta_0 &= 376.73031346177 \Omega. \end{aligned}$$

Faraday's law and Ampere's law with an impressed current $\mathbf{J}$ are

$$\nabla \times \mathbf{E} = -j\omega\mu \mathbf{H}, \tag{1}$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon \mathbf{E} + \mathbf{J}. \tag{2}$$

A real conductivity is not a second impressed current. In FDFD it enters as $\varepsilon + \sigma/(j\omega)$ when ε is real. A law that already returns a complex $\varepsilon(\omega)$ has included its loss, and the accompanying σ must be zero.

The grid lies in the xy -plane and nothing depends on z . TMz keeps E_z , H_x , and H_y . TEz keeps H_z , E_x , and E_y . The code never forms the scalar Helmholtz operator. Both drivers

eliminate, or update, the magnetic field through the curls, so a face-by-face μ stays in the magnetic samples.

On a PEC wall the tangential electric field is the boundary value stored with the grid. Milestone 1 sets that value to zero. The slab and the cylinder instead prescribe the analytic tangential trace and leave the impressed current at zero. A periodic wall identifies opposite faces with phase one. YeeGrid2D accepts `Boundary.PEC` and `Boundary.PERIODIC` only. A PML is the stretch of Sec. 3.5 on a band inside that PEC wall. A resistive or GSTC sheet remains in the formulation note. An embedded PMC is a mask on magnetic samples inside the grid. The outer wall is still `Boundary.PEC` or `Boundary.PERIODIC`.

2.1 Staircase media

A sample is inside a region when its own coordinate is inside. Electric and magnetic samples are tested separately. There is no volume fraction and no conformal cut. A slab perpendicular to y occupies the half-open interval $y_0 \leq y < y_1$. A disk of radius R centered at (x_c, y_c) occupies $(x - x_c)^2 + (y - y_c)^2 \leq R^2$. Later regions overwrite earlier ones. The background is the law passed to `StaircaseIsotropic`, usually `UniformIsotropic` with vacuum values.

2.2 Bulk dispersion

One isotropic electric pole may fill the whole domain. The staircase does not yet vary the pole in space, and there is no magnetic dispersion.

The Debye permittivity and its auxiliary are

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \Delta \varepsilon}{1 + j\omega\tau}, \quad (3)$$

$$\tau \partial_t P + P = \varepsilon_0 \Delta \varepsilon E. \quad (4)$$

The Lorentz permittivity and its auxiliary are

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \omega_p^2}{\omega_0^2 - \omega^2 + j\gamma\omega}, \quad (5)$$

$$\partial_t^2 P + \gamma \partial_t P + \omega_0^2 P = \varepsilon_0 \omega_p^2 E. \quad (6)$$

These are bulk oscillators. They are not the surface oscillators in the GSTC section of the formulation note. Ampere's update subtracts the polarization current J_p with the impressed current,

$$\mathbf{E} \leftarrow C_a \mathbf{E} + C_b (\nabla \times \mathbf{H} - \mathbf{J}_p - \mathbf{J}). \quad (7)$$

C_a and C_b are the usual lossy coefficients built on ε_∞ and σ . The Lorentz auxiliary stores P at integer steps and advances it explicitly from E^n . The Debye auxiliary is centered, $E^{n+1/2} = (E^{n+1} + E^n)/2$, and the update is local to each sample. TEz keeps one auxiliary on E_x and one on E_y . A complex ε is rejected by the leapfrog. Passing ω into `sample_tm` or `sample_te` is how FDFD sees (3) or (5) directly.

3 Discrete method

3.1 Grid

The domain is $[0, n_x \Delta x] \times [0, n_y \Delta y]$. Array axis 0 is x and axis 1 is y , so a sample is indexed $[i, j]$. Locations are those of the Yee cell. For a PEC grid the boundary electric samples are stored. TMz places E_z on nodes, H_x on the y -edges, and H_y on the x -edges. TEz places H_z at cell centers, E_x on y -edges, and E_y on x -edges. E_x is zero on $y \in \{0, b\}$ and E_y is zero on $x \in \{0, a\}$

unless a Dirichlet trace replaces those zeros. Periodic identification drops the duplicate far side, and every stored sample is an unknown.

`YeeGrid2D.coordinates` returns the 1D sample coordinates of one component. Those coordinates are the positions the staircase tests.

3.2 Curls

The discrete operators are centered differences. On the PEC grids they are transposes of each other once the boundary electric samples have been removed from the unknown vector. An invalid degree-of-freedom id is -1 and must be masked before any component offset is added. TMz uses

$$(\nabla \times E)_x = \partial_y E_z, \quad (\nabla \times E)_y = -\partial_x E_z, \quad (8)$$

and the matching Ampere pair. TEz uses

$$\nabla \times \mathbf{E} = \partial_x E_y - \partial_y E_x, \quad E_x = \partial_y H_z, \quad E_y = -\partial_x H_z. \quad (9)$$

The array curls and the sparse curls implement the same stencil. The array form keeps every sample, including the boundary values. The sparse pair used by FDFD drops the PEC rows.

The sign in (1) is not stored inside the factored FDFD matrix. Reconstructing a physical magnetic field from a Yee electric field requires

$$\mathbf{H} = -\frac{\nabla \times \mathbf{E}}{j\omega\mu}. \quad (10)$$

Omitting the minus negates H_x or H_z . The example plots use (10).

3.3 Leapfrog

One step advances $\mathbf{H}$ from time $n - \frac{1}{2}$ to $n + \frac{1}{2}$, then $\mathbf{E}$ from n to $n + 1$. With real ε and σ ,

$$C_a = \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)}, \quad C_b = \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)}. \quad (11)$$

$J = 0$ and $J_p = 0$ is the milestone-1 update. The stable step on a uniform medium of speed c satisfies

$$\Delta t \leq \frac{1}{c\sqrt{\Delta x^{-2} + \Delta y^{-2}}}. \quad (12)$$

When $\sigma = 0$ the leapfrog invariant, not the sum of the staggered sample energies, is the conserved quantity. The fully discrete frequency of a mode lies slightly above its semi-discrete frequency.

3.4 FDFD and the Dirichlet trace

Eliminating $\mathbf{H}$ produces the electric system

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) - \omega^2 \varepsilon \mathbf{E} = -j\omega \mathbf{J}, \quad (13)$$

stored as $Ae = -j$. The real eigenproblem $Ke = \omega^2 Me$ still requires real positive ε and μ and $\sigma = 0$. On a TEz PEC box the curl-curl operator has a gradient kernel, so a computed ω at roundoff is discarded before the physical spectrum is read.

Boundary electric samples stay out of the unknown vector. A trace e_D that equals the prescribed boundary values, and is zero on the free samples, moves to the right-hand side,

$$A_{ff} e_f = -j - (Ae_D)_f. \quad (14)$$

The free-row action is the array form of the left-hand side of the FDFD equation. Calling `solve` with no trace, or with a zero trace, is the milestone-1 solve. The slab and the cylinder pass `dirichlet` set to the analytic boundary field and set `j = 0`. This closes the box with a prescribed trace. The slab and the cylinder use that trace. The stretch that absorbs an outgoing wave is the next subsection.

3.5 Convolutional PML

An open side is a band of cells inside the PEC grid. The outer sample stays on the PEC wall, and depth ρ is measured from the inner face of the band toward that wall. The face itself, $\rho = 0$, is unstretched. On an axis w ,

$$s_w = \kappa_w + \frac{\sigma_w}{\alpha_w + j\omega\varepsilon_0}, \quad (15)$$

and the centered difference that approximates ∂_w is multiplied by s_w^{-1} at the sample where that difference sits. Electric and magnetic samples lie half a cell apart, so the two curls carry different scales and the stretched pair is not a transpose. With every thickness zero the scale is absent, and the curls remain the real transpose of Sec. 3.2.

Inside a band of thickness d , with polynomial order m ,

$$\sigma = \sigma_{\max} \left(\frac{\rho}{d} \right)^m, \quad \sigma_{\max} = -\frac{(m+1) \ln R(0)}{2\eta_0 d}, \quad (16)$$

together with $\kappa = 1 + (\kappa_{\max} - 1)(\rho/d)^m$ and $\alpha = \alpha_{\max}(1 - \rho/d)^m$. The defaults are $m = 3$, $R(0) = 10^{-6}$, $\kappa_{\max} = 1$, and $\alpha_{\max} = 0$. Each side uses its own thickness, the cell count times the spacing on that axis. A node on the PEC wall has $\rho = d$, so $\sigma = \sigma_{\max}$, $\kappa = \kappa_{\max}$, and $\alpha = 0$. `PMLSpec.box` puts the same cell count on all four sides. A periodic grid is rejected: the band needs the PEC backing.

FDFD rebuilds the sparse curls at the solve frequency, since s_w depends on ω . The eigenproblem keeps the unstretched real curl, and `spatial_operator` refuses an active PML.

The leapfrog stores one auxiliary per stretched derivative. It advances that auxiliary with the exponential integrator of $1/s_w$, on the same Δt as the field step. C_a and C_b stay on the material samples. A Debye or Lorentz callback still receives $\nabla \times \mathbf{H}$ after the magnetic update, and that curl is the stretched one. With no PML object, both drivers stay on the milestone-1 update.

3.6 Embedded conductors

PEC and PMC are sample masks. Setting ε to infinity, or μ to zero, is a different model. PEC removes every electric sample whose own coordinate lies in the object. PMC removes every magnetic sample in its region. The leapfrog writes those samples back to zero: PEC electric samples together with the outer wall, and PMC magnetic samples after the magnetic update, before Ampere's law reads them. The eigenproblem uses that reduced real curl. It still refuses an active PML.

The regions use the staircase tests of Sec. 2.1. A band is the half-open interval $y_0 \leq y < y_1$. A circle is closed, so a sample on the circle is inside. The outer wall stays `Boundary.PEC` or `Boundary.PERIODIC`.

The open PEC cylinder solves for the scattered field on the PML of Sec. 3.5. The trace in (14) equals $-E_{\text{inc}}$ on the object and zero on the outer wall. A free sample of the total field is $E_s + E_{\text{inc}}$. Inside the object the total electric field is zero. The TEz incident wave is $H_z = e^{-jk(x-x_c)}$, with $E_x = 0$ and $E_y = (k/\omega\varepsilon_0) H_z$.

3.7 Analytic checks

Both scattering checks use the trace (14) on every PEC electric sample. The interior material is the staircase of Sec. 2.1. The comparison is the gap left by numerical dispersion and, for the cylinder, by the staircase.

Fresnel slab. A non-magnetic slab $y_1 \leq y < y_2$ of index $n = \sqrt{\varepsilon_r}$ sits in vacuum. The incident wave travels toward $+y$ and has phase zero at the front face. TMz reflects with $r = (1-n)/(1+n)$. TEz reflects H_z with $r_H = (n-1)/(n+1) = -r$, so $R_H = -R$ while the transmission amplitude, including the transit phase e^{-jkd} , is the same. The interior amplitudes are not the same: TMz splits $1 \pm R$ with $1/n$, and TEz splits $1 \pm R_H$ with n . When $\varepsilon_r = 1$, both reflections vanish and the field is the incident wave, not a transmission of 1. The closed forms are `slab_coefficients` and `slab_fields`. A forced branch (`below`, `slab`, `above`) evaluates one piecewise piece at a face, where the two neighboring expressions must agree.

The comparison wavelength is kept off the TEz eigenfrequencies of the PEC box. On the 8×80 domain with $\Delta = 1$, $\lambda = 40$ is one of those frequencies and the Dirichlet matrix is singular. The shipped example uses $\lambda = 46$, $\Delta = 0.5$, and faces $y = 30.375$ and $y = 50.375$, which miss the electric nodes of both that grid and the once-refined grid.

Dielectric cylinder. The outgoing wave is $H_n^{(2)}$. The incident field e^{-jkx} has the Jacobi–Anger coefficients $(-j)^n$. TMz is the E_z series. TEz is the H_z series, with the contrast on the undifferentiated interior Bessel function, because E_φ brings $1/\varepsilon$. The PEC coefficient $a_n = -(-j)^n J_n(ka)/H_n^{(2)}(ka)$ is a separate regression. It is not the limit $\varepsilon_r \rightarrow \infty$ of the dielectric formula. The TEz form, from $E_\varphi = 0$, is $a_n = -(-j)^n J'_n(ka)/H_n^{(2)'}(ka)$. The truncation follows the MaxwellMOM rule $\lceil |ka| + 4|ka|^{1/3} + 8 \rceil$, and the derivative is the recurrence $\frac{1}{2}(f_{n-1} - f_{n+1})$.

The open dielectric cylinder compares this series on the PML of Sec. 3.5. The unknown is the scattered field and the outer wall is homogeneous. The incident wave enters as the impressed current $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$, sampled with the staircase permittivity that stays in the operator. A free sample of the total field is $E_s + E_{\text{inc}}$. The closed-box cylinder still prescribes the Mie trace.

Line current. A z -directed current whose integral is I satisfies $(\nabla^2 + k^2)E_z = j\omega\mu J_z$ in a uniform medium. For $e^{+j\omega t}$ the outgoing field is $E_z = -(\omega\mu I/4) H_0^{(2)}(k\rho)$, with $k = \omega\sqrt{\mu\varepsilon}$ taken from the same μ and ε the solver uses. The discrete sample is $J_z = I/(\Delta x \Delta y)$. The comparison keeps the unstretched samples at least $\lambda/2$ from the source. The continuous field diverges at $\rho = 0$, and that sample is left undefined.

3.8 3D cell

The 3D grid stores all six components. E_x sits at $((i + \frac{1}{2})\Delta x, j\Delta y, k\Delta z)$, and the other electric components are staggered on their own axes. PEC drops the tangential electric samples. Magnetic samples that do not touch a free electric sample are omitted, so the unstretched curls remain transposes. The FDFD matrix is the elimination in (13), with the diagonal $\varepsilon + \sigma/(j\omega)$ when ε is real. The rectangular cavity frequency is the two-dimensional semi-discrete formula with the added term $(2 \sin(k_z \Delta z/2)/\Delta z)^2$ and $k_z = p\pi/c$. A mode with exactly one index equal to zero is one Cartesian component: $(m, n, 0)$ is E_z , $(m, 0, p)$ is E_y , and $(0, n, p)$ is E_x .

4 Code

The installable package is `maxwell_fd`, version 0.1.0. Its source is `src/maxwell_fd`. NumPy and SciPy are required. Matplotlib is the `viz` extra and is also a dependency of the conda environment `maxwell-fd`.

Module	Role
<code>grid.yee2d</code>	YeeGrid2D, Polarization, Boundary, coordinates
<code>materials.volume</code>	Uniform, staircase, mapped $\varepsilon(x, y)$, component arrays
<code>materials.dispersion</code>	DebyeDielectric, LorentzDielectric, and their auxiliaries
<code>materials.conductors</code>	Staircase PEC and PMC masks, YBand and Circle
<code>operators.curl2d</code>	Sparse <code>curl_e</code> , <code>curl_h</code> , array curls, DOF layout, optional stretch
<code>operators.pml</code>	CFS profile, $1/s_w(\omega)$, and the leapfrog auxiliaries
<code>drivers.fdt2d</code>	<code>step_tmz</code> , <code>step_tez</code> , CFL step, leapfrog invariant
<code>drivers.fdfd2d</code>	FDFDOperator: system, eigenproblem, Dirichlet solve
<code>analytics.cavity2d</code>	Semi-discrete cavity frequencies and mode shapes
<code>analytics.dispersion</code>	Fully discrete frequency and the three-sample fit
<code>analytics.fresnel</code>	Slab coefficients and fields
<code>analytics.mie2d</code>	Cylinder coefficients and E_z , H_z , E_x , E_y
<code>analytics.green2d</code>	Outgoing line-current E_z
<code>grid.yee3d</code>	YeeGrid3D: six components, PEC and periodic shapes
<code>materials.volume3d</code>	Uniform isotropic ε , μ , and σ
<code>operators.curl3d</code>	Sparse and array curls on the 3D cell
<code>drivers.fdfd3d</code>	FDFDOperator3D: system and real eigenproblem
<code>analytics.cavity3d</code>	Semi-discrete frequency with the z term, and the pure modes

A frequency-domain TMz solve with a staircase disk has this shape. The trace argument is omitted when the wall is a homogeneous PEC.

```

grid = YeeGrid2D(nx, ny, dx, dy, Polarization.TMZ, Boundary.PEC)
law = StaircaseIsotropic(
    UniformIsotropic(),
    (Disk(cx, cy, radius, eps=eps_r * EPS0),),
)
operator = FDFDOperator(grid, law.sample_tm(grid))
ez = operator.solve(omega, np.zeros(operator.layout.n_e),
    dirichlet={"ez": ez_trace})

```

`sample_tm(grid)` and `sample_te(grid)` without ω return the real instantaneous coefficients used by the leapfrog (ε_∞ when the law is dispersive). Passing ω allows a complex permittivity. `step_tmz` and `step_tez` take the impressed current and the polarization current as optional arguments. Their defaults reproduce the milestone-1 update. The same two steps take an optional `ConvolutionalPML` and an optional `Conductors` mask. `FDFDOperator` takes an optional `PMLSpec` and the same mask. Passing neither a PML nor a conductor leaves both drivers on the full unstretched curls.

The real eigenproblem is `spatial_operator`. It returns the pair (K, M) of the unstretched curl and refuses an active PML. TEz results should drop nonpositive eigenvalues before they are compared with the semi-discrete formula.

5 Examples

Scripts under `examples/fdfd/2d/`, `examples/fdfd/3d/`, and `examples/fdtd/` are the runnable demos. The time-domain tree does not yet solve a scattering problem. From the project directory, with the conda environment `maxwell-fd` active,

```

python examples/fdfd/2d/fresnel_slab.py
python examples/fdfd/2d/dielectric_cylinder.py
python examples/fdfd/2d/pml_line.py

```

```
python examples/fdfd/2d/pec_cylinder.py
python examples/fdfd/2d/dielectric_cylinder_open.py
python examples/fdfd/3d/cavity.py
```

Each script writes a sibling directory whose name is the script stem plus the parameters of that run, in the order they are passed to `examples/run_dir.output_dir`. Floats are formatted with twelve significant digits and trailing zeros removed, so 2.0 becomes 2 and 0.5 stays 0.5. Keys do not contain underscores.

Field maps use `pcolormesh` with Gouraud shading. Each polarization is a row of three columns: the Yee field, the analytic field, and their difference. The cylinder columns are FD, Mie, and FD–Mie. The slab columns are FD, Fresnel, and FD–Fresnel. The line-current columns are FDFD, Hankel, and FDFD–Hankel. Color limits of the first two columns match. The difference has its own scale. The cylinder outline, the slab faces, or the inner edge of the PML, is drawn on every panel. The open PEC cylinder and the open dielectric cylinder draw the outline and that inner edge together. The 3D cavity figure is one panel per pure mode.

5.1 Fresnel slab

```
python examples/fdfd/2d/fresnel_slab.py \
    --index 2 --thickness 20 --y1 30.375 --wavelength 46 \
    --length 8 --height 80 --dx 0.5 --pml 1
```

`-index` is the refractive index n , and the permittivity used in the solve is $\varepsilon_r = n^2$. The default $n = 2$ is $\varepsilon_r = 4$. `-length` and `-height` are the region sides along x and y . Both must be a positive integer number of cells of size `-dx`, and each direction needs at least two cells. `-pml` is a thickness at each end in y only. It must be a positive multiple of Δ and must leave the slab in the open interval $(\text{pml}, \text{height} - \text{pml})$.

The default run writes

```
fresnel_slab_index2_d20_lam46_y130.375_length8_height80_dx0.5_pml1/
```

- `domain.png`, the mesh, the slab, the y -end bands, and the x walls;
- `fresnel_slab.png`, TMz E_z and TEz H_z in three columns;
- `cuts.png`, the center-column cuts of E_z , H_x , E_x , and H_z against the analytic curves.

On this grid the free-sample electric L^2 errors annotated on the figures are 0.23% (TMz) and 1.30% (TEz). Those numbers are the result of this particular grid. The acceptance bars in the formulation note are wider and were not changed to match them.

Propagation is $+y$, drawn horizontal so the figure reads with the line cuts. The incident field does not depend on x , and neither does the slab. The orange bands are therefore only the two ends $y = 0$ and $y = b$. The walls $x = 0$ and $x = a$ are drawn in green and labeled as the PEC trace: every boundary electric sample on those walls is set to the analytic Fresnel value, together with the samples on $y = 0$ and $y = b$. The slab rectangle spans the full width in x . Figure 1 is that layout. The orange bands mark a drawn thickness at $y = 0$ and $y = b$. `FDFDOperator` is constructed here without a `PMLSpec`, so every derivative stays the centered difference of Sec. 3.2 and the trace is the analytic field. The stretch itself is the line-current example of Sec. 5.3.

5.2 Dielectric cylinder

```
python examples/fdfd/2d/dielectric_cylinder.py \
    --radius 10 --index 1.5 --size 50 --dx 1 --pml 8
```

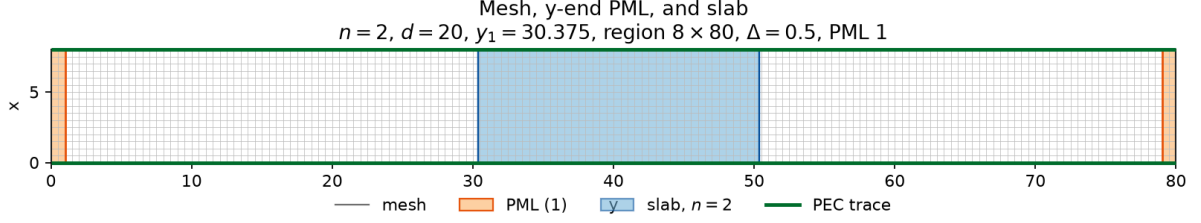


Figure 1: Default Fresnel domain. Orange bands are the y -end PML thickness drawn on the figure. Green edges are the PEC trace at $x = 0$ and $x = 8$.

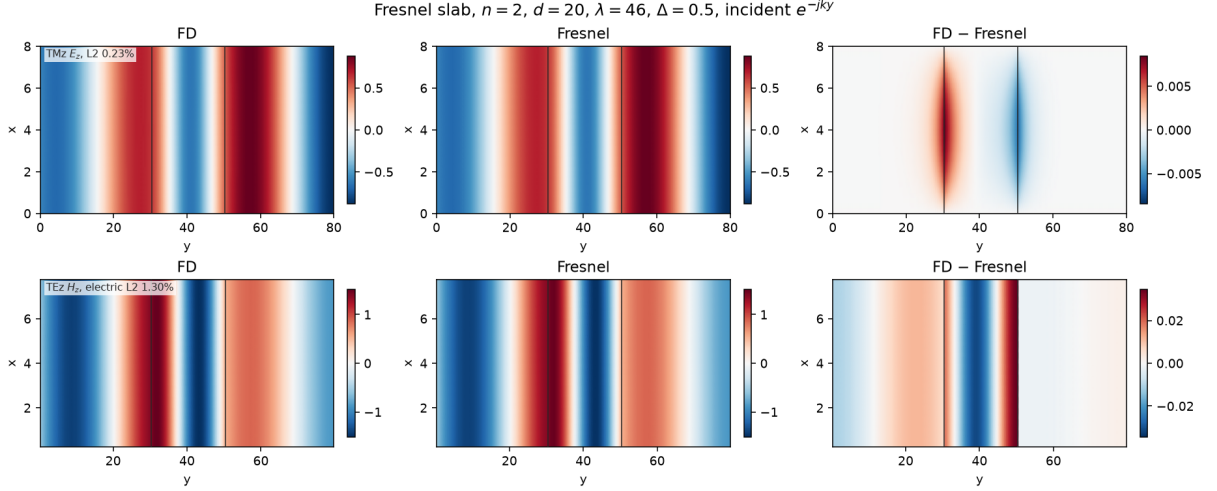


Figure 2: Default Fresnel maps. Vertical lines mark the slab faces. H_z is rebuilt with (10). The annotated L^2 errors are the free electric samples.

The region is a square of side `-size`. `-index` is again n , with $\varepsilon_r = n^2$. The default $n = 1.5$ is $\varepsilon_r = 2.25$. The size parameter ka is held at 1, so $k = 1/a$ and the wavelength scales with the radius. The radius must be positive and smaller than half the side length minus the PML thickness. The default run writes

```
dielectric_cylinder_index1.5_ka1_a10_size50_dx1_pml8/
```

with `domain.png` and `dielectric_cylinder.png`.

The cylinder scatters in the plane of the grid, so the domain figure draws the PML thickness on all four sides. As in the slab example, that frame is a layout. The solve is still (14) on the PEC wall, with the Mie field as the trace and with the staircase disk as the only material contrast. Every derivative in this solve is the centered difference of Sec. 3.2. TMz shows $\text{Re } E_z$. TEz shows $\text{Re } H_z$ from (10), and the annotated L^2 error is the free-sample error in E_x and E_y . On the default grid those errors are 0.44% and 2.07%.

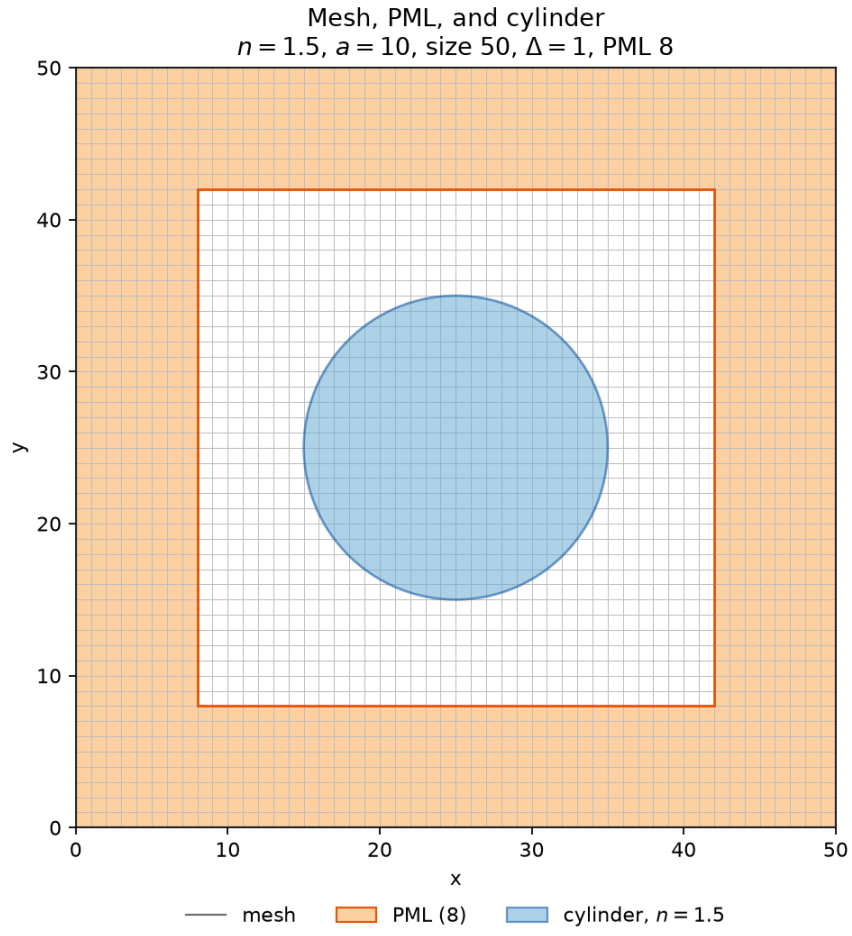


Figure 3: Default cylinder domain. The orange frame is the drawn PML thickness. The disk is the staircase geometry. The outer boundary of the solve remains the PEC trace.

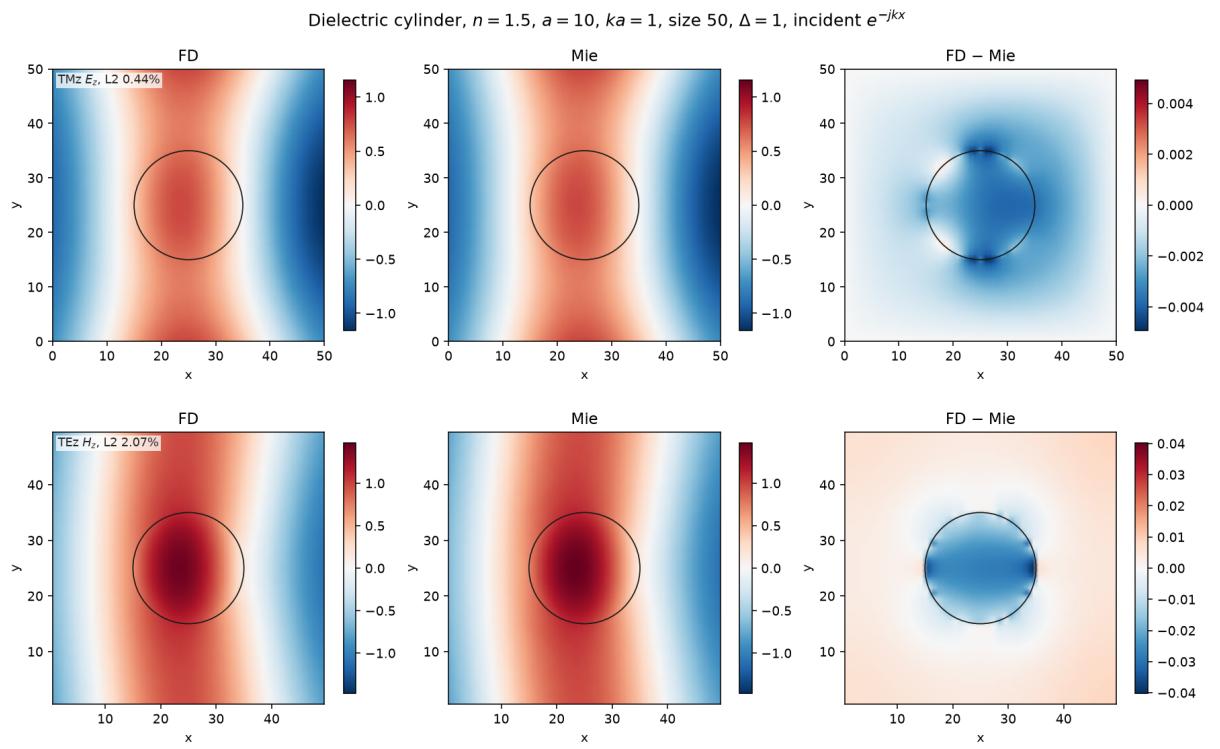


Figure 4: Default cylinder maps against the Mie series at $ka = 1$, $n = 1.5$.

5.3 Line current in a PML

```
python examples/fdfd/2d/pml_line.py \  
    --wavelength 1 --side 3 --dx 0.05 --pml 8
```

The domain is a square PEC box of side `-side`. `-dx` is the cell size, and `-side` must be an integer number of those cells, at least two. `-pml` is the cell count on every side. It is at least one, and twice that count stays below the number of cells, so an interior remains. `-wavelength` is the free-space wavelength. The source is the center E_z sample, $J_z = 1/(\Delta x \Delta y)$, and the reference is the line current of Sec. 3.7 with $I = 1$. The default run writes

pml_line_lam1_side3_dx0.05_pml8/

and `pml_line.png`. The orange square is the inner edge of the band. On this grid the interior L^2 error, outside half a wavelength of the source, is 2.10%. That number belongs to this grid. The Hankel panel is the unbounded cylindrical wave, including through the band, so the difference outside the square is the absorption.

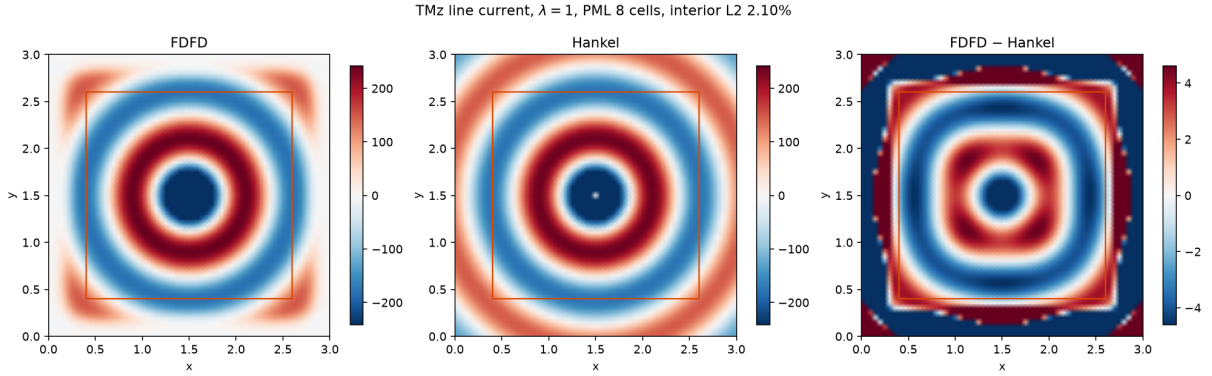


Figure 5: Line current at $\lambda = 1$ on a 3 m box, $\Delta = 0.05$ m, with eight PML cells on each side. The orange square is the inner edge of the band. The annotated error is the interior window described above.

5.4 Open PEC cylinder

```
python examples/fdfd/2d/pec_cylinder.py \
    --radius 8 --size 64 --dx 1 --pml 8
```

The region is a square of side `-size`. `-pml` is the cell count of the convolutional PML on each side, the stretch of Sec. 3.5. ka stays at 1. The solve frequency is $\omega = c_0/a$, and the Mie series uses $k = \omega\sqrt{\mu_0\epsilon_0}$. The radius must lie inside the unstretched square. The default run writes

`pec_cylinder_ka1_a8_size64_dx1_pml8/`

with `domain.png` and `pec_cylinder.png`.

The unknown is the scattered field. The cylinder trace is $-E_{\text{inc}}$ and the outer wall is zero. A free sample of the total field is $E_s + E_{\text{inc}}$. TMz shows $\text{Re } E_z$. TEz shows $\text{Re } H_z$ from (10). The annotated L^2 is the electric-sample error on the unstretched samples at least 2Δ outside the circle. On this grid those errors are 4.20% (TMz) and 5.58% (TEz). The orange square is the inner edge of the PML. Outside that square the difference is the absorption together with the scattered field's wall condition.

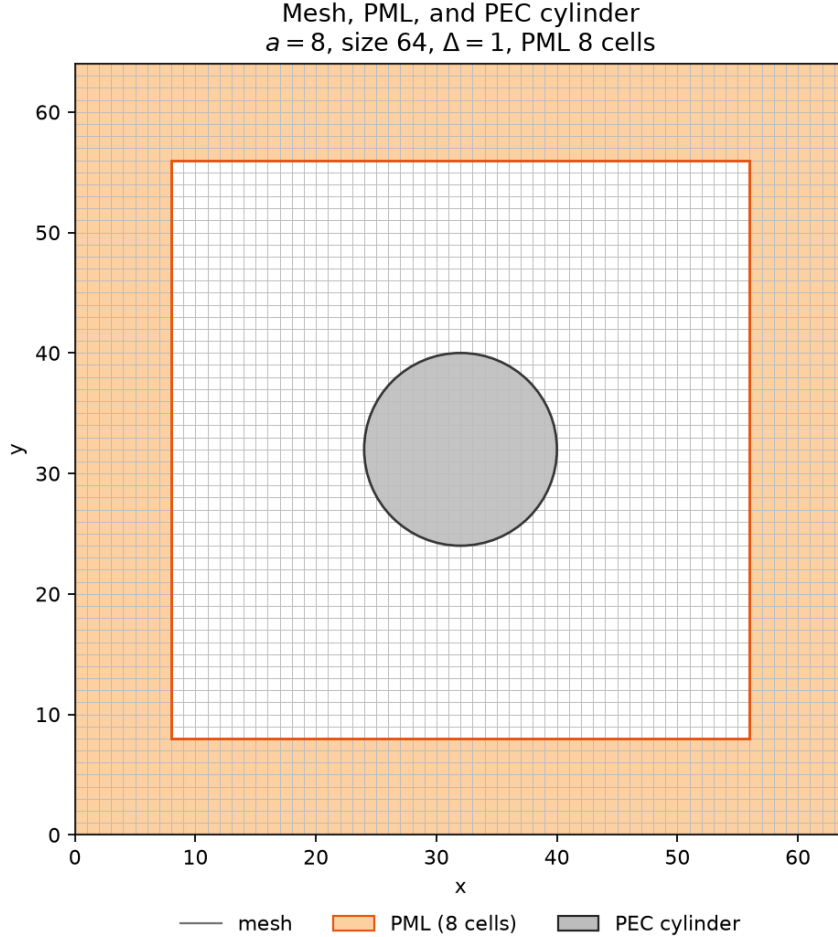


Figure 6: Default open-cylinder domain. The orange frame is the PML band used by the solve. The disk is the staircase PEC.

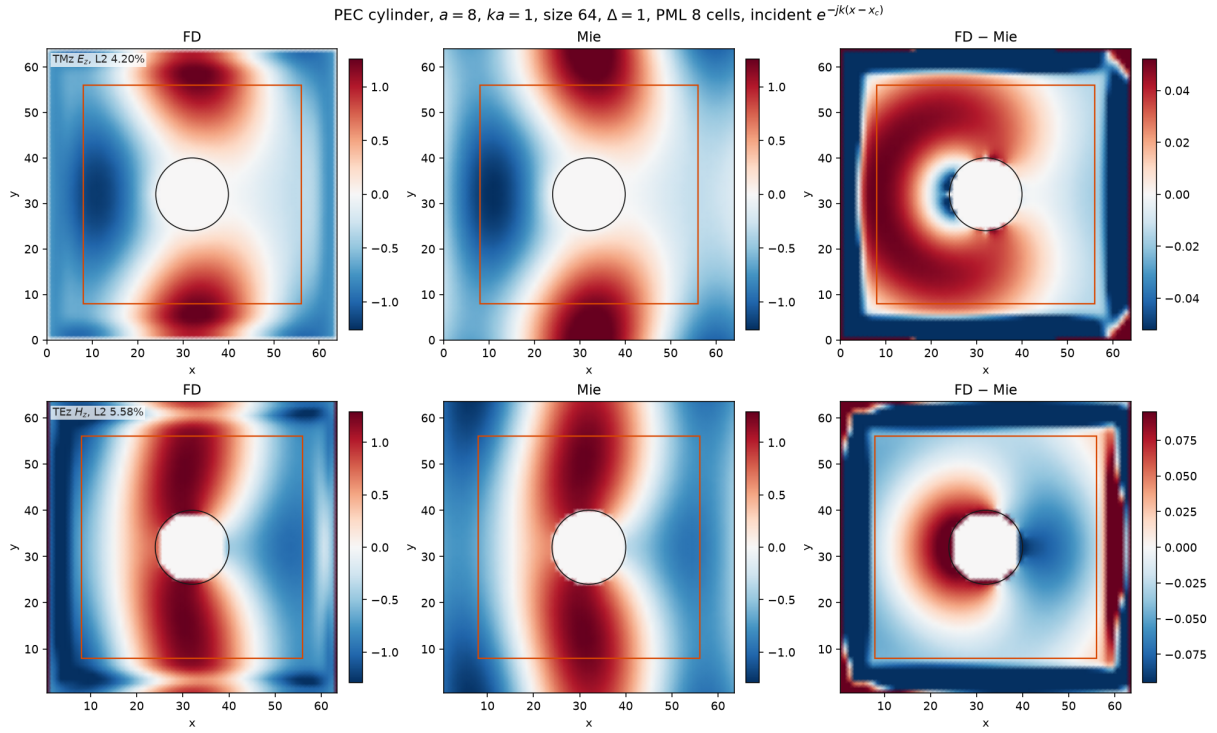


Figure 7: Open PEC cylinder at $ka = 1$, $a = 8$, $\Delta = 1$, with eight PML cells on each side. The annotated L^2 errors are the electric samples in the unstretched window, at least 2Δ outside the circle.

5.5 Open dielectric cylinder

```
python examples/fdfd/2d/dielectric_cylinder_open.py \
    --radius 8 --index 1.5 --size 64 --dx 1 --pml 8
```

The region is a square of side `-size`. `-index` is n , with $\varepsilon_r = n^2$. The default $n = 1.5$ is $\varepsilon_r = 2.25$. `-pml` is the cell count of the convolutional PML on each side, the stretch of Sec. 3.5. ka stays at 1. The solve frequency is $\omega = (ka/a) c_0$, and the Mie series uses $k = \omega \sqrt{\mu_0 \varepsilon_0}$. The radius must lie inside the unstretched square. The default run writes

```
dielectric_cylinder_open_index1.5_ka1_a8_size64_dx1_pml8/
```

with `domain.png` and `dielectric_cylinder_open.png`.

The unknown is the scattered field. The operator holds the staircase disk. The impressed current is $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$ on the electric samples, and it is zero in the vacuum. The outer wall is a homogeneous PEC. A free sample of the total field is $E_s + E_{\text{inc}}$. TMz shows $\text{Re } E_z$. TEz shows $\text{Re } H_z$ from (10). The annotated L^2 is the electric-sample error on the unstretched samples at least 2Δ off the circle, inside and outside. On this grid those errors are 0.45% (TMz) and 0.71% (TEz). The orange square is the inner edge of the PML. The closed-box cylinder of Sec. 5.2 still prescribes the Mie trace and does not stretch the derivatives.

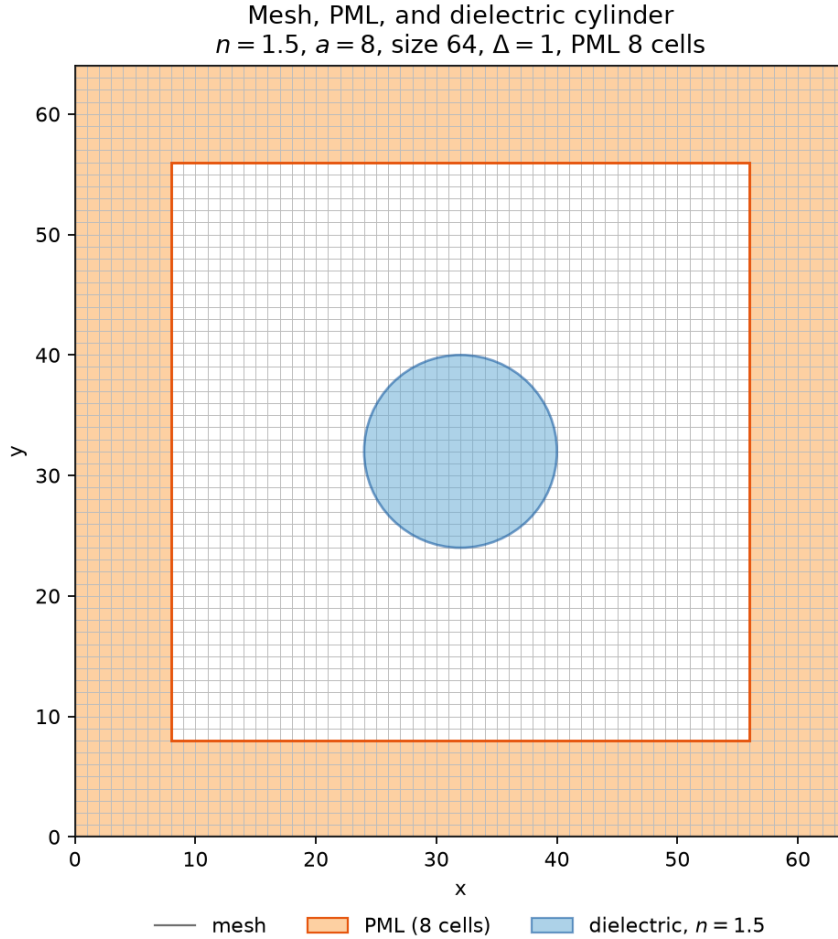


Figure 8: Default open dielectric domain. The orange frame is the PML band used by the solve. The disk is the staircase dielectric, $n = 1.5$.

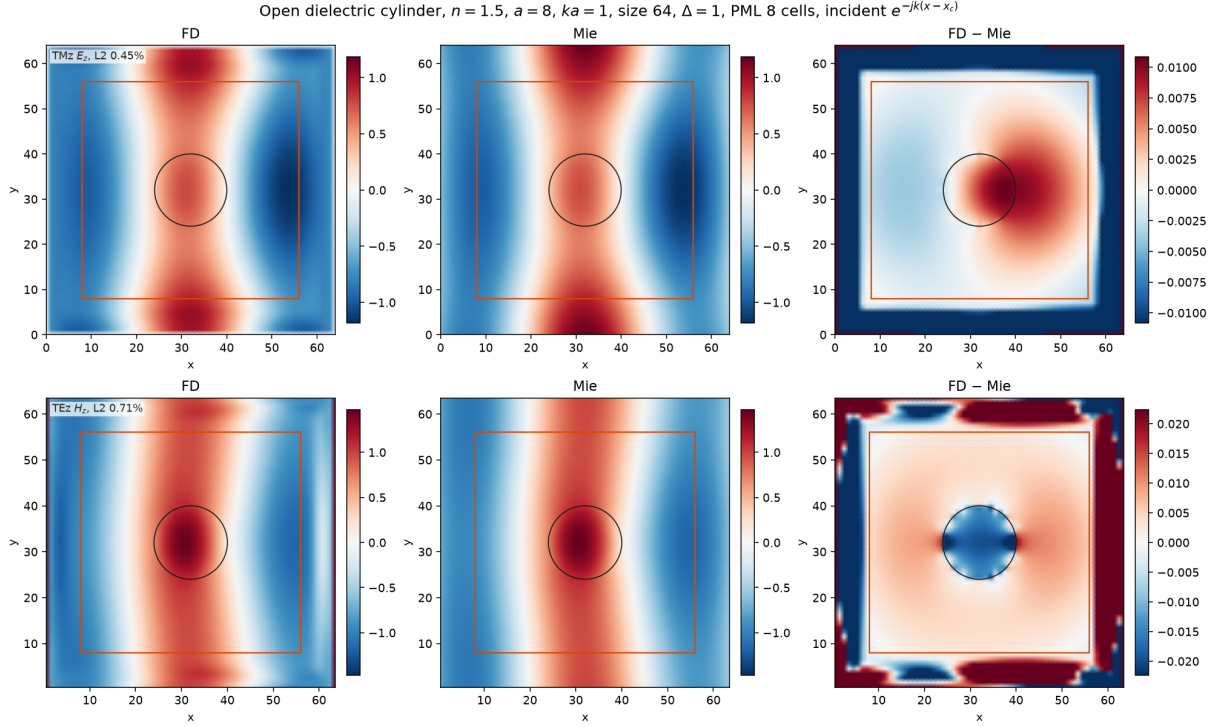


Figure 9: Open dielectric cylinder at $n = 1.5$, $ka = 1$, $a = 8$, $\Delta = 1$, with eight PML cells on each side. The annotated L^2 errors are the electric samples in the unstretched window, at least 2Δ off the circle.

5.6 3D PEC cavity

```
python examples/fdfd/3d/cavity.py --nx 16 --ny 12 --nz 10 --dx 1
```

The box is a PEC cavity with one spacing on every axis. The default run writes

`cavity_nx16_ny12_nz10_dx1/`

with `cavity.png`. The three panels are the pure modes $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$. Each panel shows the one nonzero electric component, on the plane in which that mode varies. The script prints the relative Rayleigh residual of each mode on that grid. The regression in the test suite uses the coarser $8 \times 6 \times 5$ box and the bar 10^{-10} .

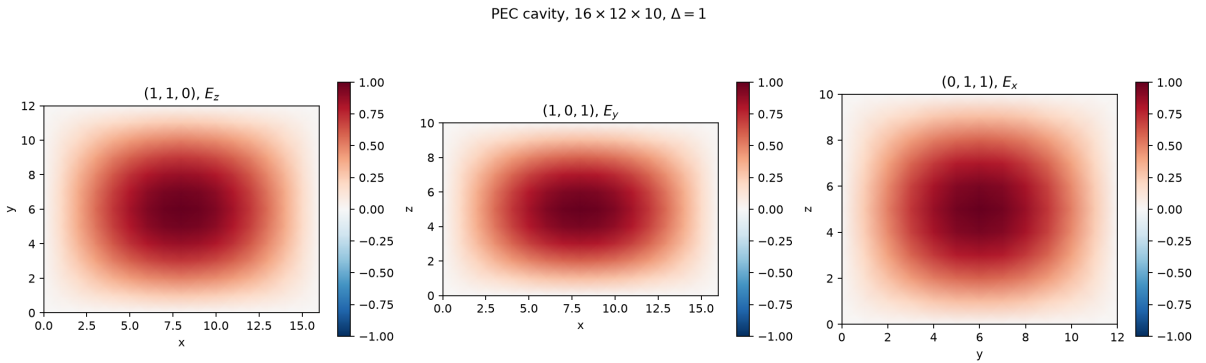


Figure 10: Pure modes of a $16 \times 12 \times 10$ PEC cavity at $\Delta = 1$. $(1, 1, 0)$ is $E_z(x, y)$, $(1, 0, 1)$ is $E_y(x, z)$, and $(0, 1, 1)$ is $E_x(y, z)$.

6 Tests and later slices

The suite is `pytest` from the project root, fifty-three tests in `tests/`. Milestone 1 checks the curls, the adjoint, the cavity eigenvalues, the manufactured solve, the invariant and the CFL growth, and the fully discrete dispersion, with relative or absolute bars from 10^{-12} to 10^{-6} as listed in the formulation note. Milestone 2 adds the manufactured solve with $\sigma > 0$ and with a smooth $\varepsilon(x, y)$, a uniform $\varepsilon_r = 4$ cavity, the slab-loaded cavity, the lossless staircase invariant, Fresnel face continuity, the Mie continuity and PEC coefficient, the Fresnel and cylinder L^2 bars of 2×10^{-2} , and one period of Debye and Lorentz evolution. The PML tests check the grading, the stretched curls, a zero-thickness step, the line current, and the leapfrog phasor. The formulation note records those bars: a coarse interior error below 4×10^{-2} , a $2.5\times$ drop when the spacing is halved on the same physical band, a bare PEC window above 0.5, and a phasor error below 1.5×10^{-2} at 48 steps per period. The conductor tests compare a PEC band with the shorter cavity it leaves, a PMC quarter-wave with the sine formula, and an open PEC cylinder with the Mie series. The eigenvalue bars are 10^{-10} . The open PEC bars are 8×10^{-2} (TMz) and 10^{-1} (TEz), on the 64^2 grid of Sec. 5.4. The open dielectric bars, at $n = 1.5$ on that same grid, are 8×10^{-3} (TMz) and 1.3×10^{-2} (TEz). That comparison keeps the unstretched samples at least 2Δ off the rim, inside and outside the disk. The 3D curl bars are 10^{-12} . On an $8 \times 6 \times 5$ PEC box the three pure modes match the semi-discrete frequency, with the z term included, to 10^{-10} . A manufactured solve on a $6 \times 5 \times 4$ box recovers the electric vector to the same bar. A failing bar is a reason to refine the grid in that test, not to widen the bar.

The formulation note already fixes the formulas for the slices that remain to be coded: a 3D sphere small enough to factor, a resistive sheet, the symmetric GSTC cell, impressed beams, a near-to-far integral, and a Bloch phase. The Fresnel and dielectric-cylinder domain figures still draw an orange thickness as a layout. Those two solves use the Dirichlet trace. The open PEC cylinder of Sec. 5.4 and the open dielectric cylinder of Sec. 5.5 are the solves that stretch the derivatives.

References

- [1] K. S. Yee, “Numerical solution of initial boundary value problems involving Maxwell’s equations in isotropic media,” *IEEE Trans. Antennas Propag.*, 1966.
- [2] T. J. Smy, S. Stewart, and S. Gupta, “Integrated generalized sheet transition conditions in a Yee-cell based FDTD simulation of electromagnetic metasurfaces,” arXiv:1706.10136, 2017.